

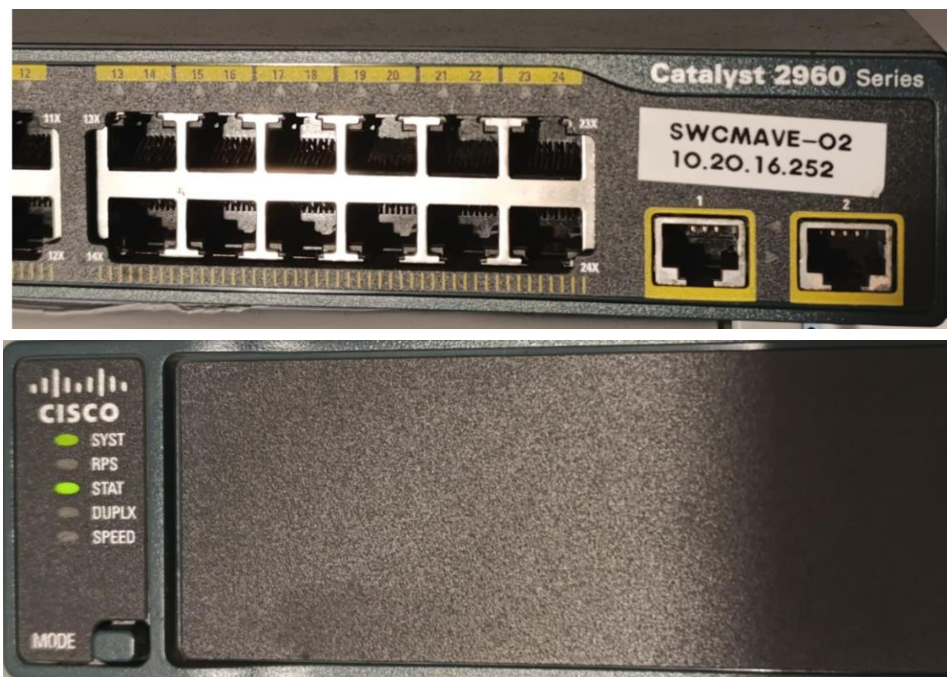
LAB NOC 01

Cisco Catalyst 2960 — Console Access & Password Recovery

Real Hardware | ES / EN | NOC Portfolio

INFORMACIÓN DEL LABORATORIO / LAB INFORMATION

Hardware / Hardware	Cisco Catalyst 2960 (WS-C2960-24TT-L)	Cisco Catalyst 2960 (WS-C2960-24TT-L)
Categoría / Category	Acceso físico y recuperación	Physical access & recovery
Nivel / Level	CCNA — Fundamentos de switching	CCNA — Switching fundamentals
Duración / Duration	Aprox. 45 minutos	Approx. 45 minutes
Entorno / Environment	Hardware real (equipo usado)	Real hardware (used equipment)
Relevancia NOC	Acceso de emergencia, troubleshooting L1	Emergency access, L1 troubleshooting



Observed indicators:

- SYST LED: solid green
- STAT LED: on
- Fan: operational

Conclusion: normal boot process

**OBJETIVO / OBJECTIVE**

OBJETIVO	OBJECTIVE
Realizar acceso por consola, recuperación de contraseña y arranque limpio en un switch Cisco Catalyst 2960 usando hardware real.	Perform console access, password recovery, and clean boot on a Cisco Catalyst 2960 using real hardware.

PROCEDIMIENTO / PROCEDURE

#	ESPAÑOL	ENGLISH
1	Conexión eléctrica segura del equipo	Safe power connection of equipment
2	Conexión de cable de consola USB–RJ45	USB–RJ45 console cable connection
3	Identificación del puerto COM en Windows (Administrador de dispositivos → COM5)	COM port identification in Windows (Device Manager → COM5)
4	Configuración de PuTTY: 9600 bps, 8N1, Flow Control: None	PuTTY configuration: 9600 bps, 8N1, Flow Control: None
5	Acceso inicial por consola (Switch>)	Initial console access (Switch>)
6	Interrupción del arranque manteniendo botón MODE durante el encendido	Boot interruption by holding MODE button during power-on
7	Inicialización del sistema de archivos flash	Flash file system initialization
8	Renombrado de config.text y vlan.dat (archivos de configuración)	Renaming config.text and vlan.dat (configuration files)
9	Arranque limpio del IOS	Clean IOS boot
10	Acceso a modo EXEC (Switch>) — configuración de fábrica verificada	Access to EXEC mode (Switch>) — factory configuration verified

TROUBLESHOOTING DOCUMENTADO / DOCUMENTED TROUBLESHOOTING

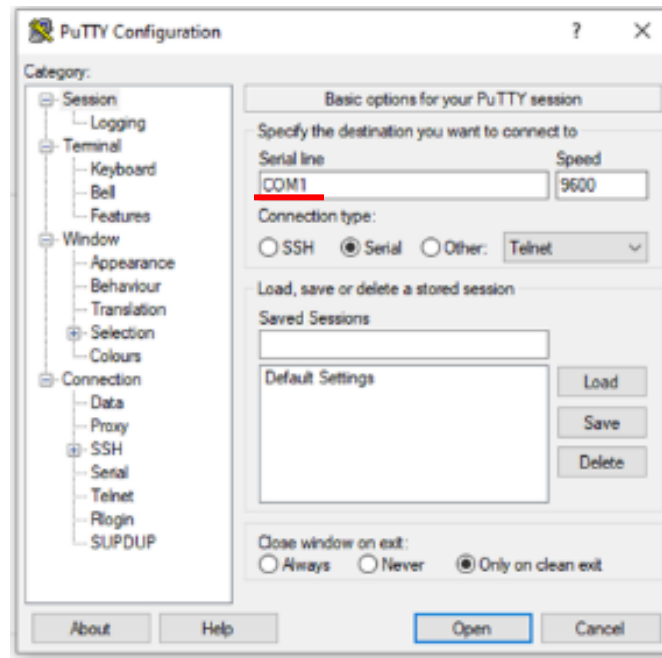
HALLAZGO / FINDING	ACCIÓN CORRECTIVA / CORRECTIVE ACTION
Puerto COM incorrecto seleccionado al inicio (COM1). / Incorrect COM port selected initially (COM1).	Puerto correcto identificado mediante Administrador de dispositivos → COM5. Parámetros de consola corregidos. / Correct port identified via Device Manager → COM5. Console parameters corrected.

Indicadores LED al encendido: SYST verde fijo, STAT encendido, ventilador operativo. / Boot LED indicators: SYST solid green, STAT on, fan operational.

Proceso de arranque normal verificado. Sin fallas de hardware. / Normal boot process verified. No hardware failures.

Issue: Incorrect COM port (COM1)

Fix: Identified correct port (COM5) via Device Manager



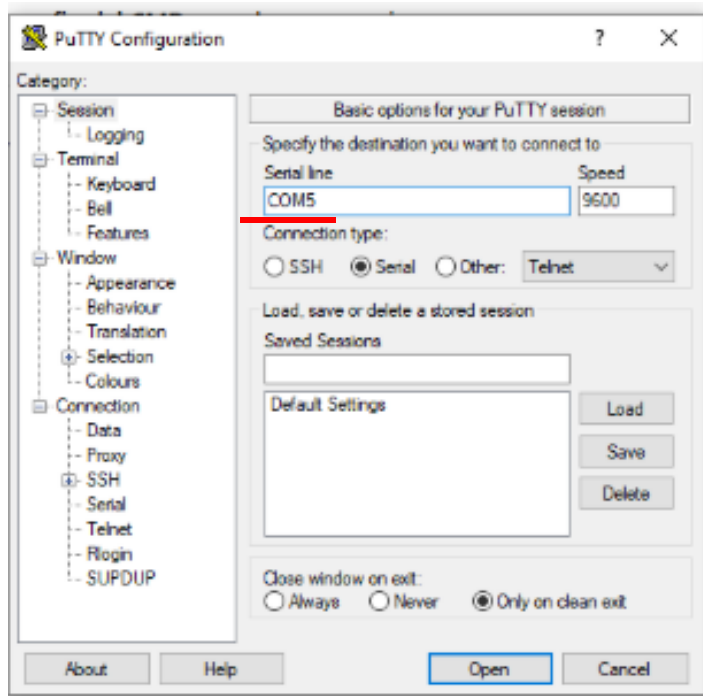
```
Microsoft Windows [Versión 10.0.19045.6466]
(c) Microsoft Corporation. Todos los derechos reservados.

C:\Users\martins>mode

Estado para dispositivo COM5:
-----
Baudios:          1200
Paridad:          Even
Bits de datos:    7
Bits de paro:     1
Tiempo de espera: OFF
XON / XOFF:       OFF
Protocolo CTS:    OFF
Protocolo DSR:    OFF
Sensibilidad de DSR: OFF
Circuito DTR:     ON
Circuito RTS:     ON

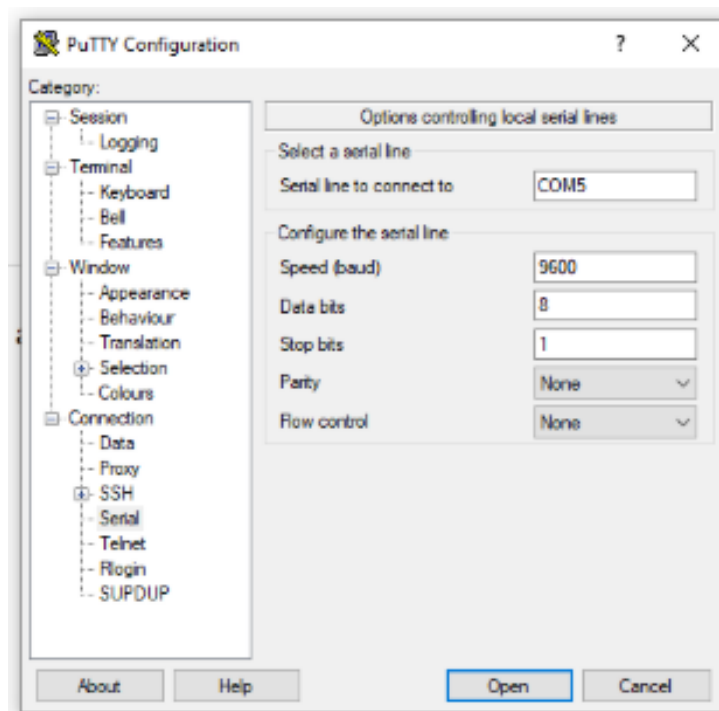
Estado para dispositivo COM1:
-----
Líneas:           9801
Columnas:         120
Ritmo del teclado: 31
Retardo del teclado: 1
Página de códigos: 850

C:\Users\martins>
```



En PuTTY → Session:

- Connection type: **Serial**
- **Serial line:** COM5
- **Speed:** 9600



PuTTY → Connection → Serial

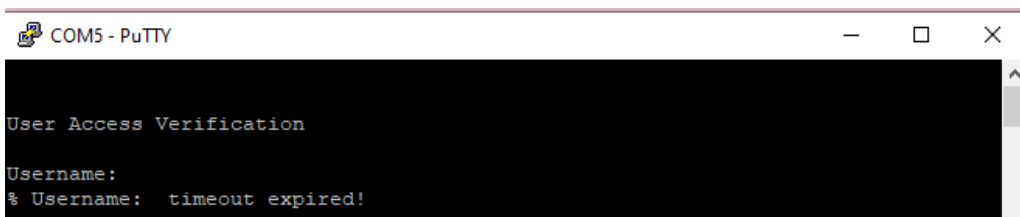
Configuration:

- Speed (baud): **9600**
- Data bits: **8**
- Stop bits: **1**
- Parity: **None**
- Flow control: **None**

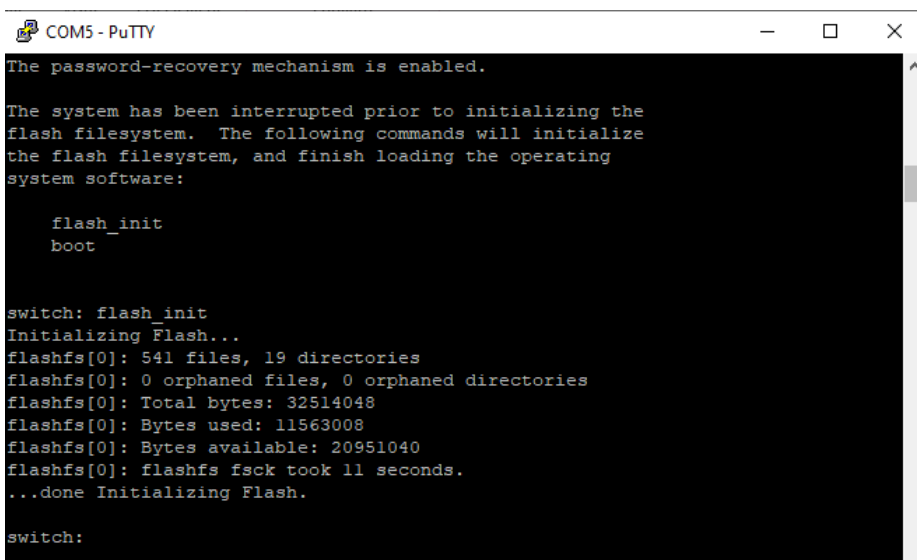
Initial Access to Cisco Catalyst 2960



```
COM5 - PuTTY
User Access Verification
Username: █
```



```
COM5 - PuTTY
User Access Verification
Username:
% Username: timeout expired!
```



```
COM5 - PuTTY
The password-recovery mechanism is enabled.

The system has been interrupted prior to initializing the
flash filesystem. The following commands will initialize
the flash filesystem, and finish loading the operating
system software:

    flash_init
    boot

switch: flash_init
Initializing Flash...
flashfs[0]: 541 files, 19 directories
flashfs[0]: 0 orphaned files, 0 orphaned directories
flashfs[0]: Total bytes: 32514048
flashfs[0]: Bytes used: 11563008
flashfs[0]: Bytes available: 20951040
flashfs[0]: flashfs fsck took 11 seconds.
...done Initializing Flash.

switch:
```

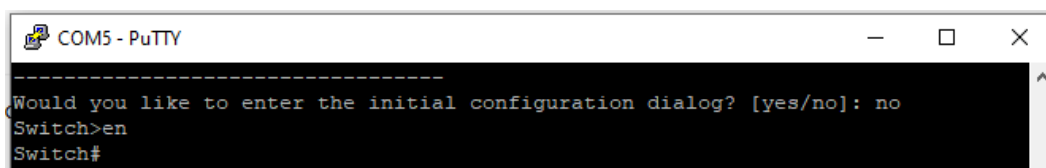
RESULTADO / RESULT

✓RESULTADO (ES)

Switch operativo con configuración de fábrica.
Recuperación de contraseña completada sin
pérdida de servicio adicional.

✓RESULT (EN)

Switch running with factory-clean configuration.
Password recovery completed without additional
service disruption.



```
COM5 - PuTTY
-----
Would you like to enter the initial configuration dialog? [yes/no]: no
Switch>en
Switch#
```

KEY TAKEAWAY

"Los cables de consola USB crean puertos COM dinámicos en el sistema operativo. Siempre verifico el puerto COM asignado antes de abrir una sesión serial, evitando pérdida de tiempo en troubleshooting."

"USB console cables create dynamic COM ports in the OS. I always verify the assigned COM port before opening a serial session — avoiding wasted troubleshooting time."

NOC RELEVANCE — MINING / INDUSTRIAL ENVIRONMENT

In a mining NOC (e.g., Newmont Cerro Negro), console access is the last resort when remote management (SSH/Telnet) is unavailable due to network failure or locked credentials. The ability to perform physical password recovery on Cisco switches — without factory reset of VLANs or trunk configs — is a critical L1/L2 skill for maintaining uptime in OT/IT converged environments.